

SIMATS ENGINEERING

ASSESSMENT TOOL 1: Data Interpretation

COURSE NAME: COMPUTER VISION

COURSE CODE: DSA0204

COURSE OUTCOME COVERED

CO4: *Compare object and pattern recognition techniques for interpreting visual data with spatial and motion characteristics. (BTL 5)*

Assessment Tool Weightage: 40%

Code of Conduct:

I __Chalamala Sai Harshitha (192424346)__ certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: C. Sai Harshitha

Read the given data carefully and compare the techniques or results based on multiple factors such as accuracy, robustness, and computational constraints. Evaluate and justify your decision with appropriate reasoning considering the application context.

1. Stereo Vision vs Motion Estimation

Depth estimation results:

- **Stereo Vision:** accurate but requires dual cameras
- **Motion Estimation:** less accurate but works with single camera

Compare both approaches and evaluate which is preferable under hardware constraints. Justify your choice.

Introduction

Depth estimation helps a computer vision system determine the distance of objects from the camera. **Stereo Vision** and **Motion Estimation** are two approaches with different advantages. The given data shows that Stereo Vision provides better accuracy, whereas Motion Estimation requires less hardware.

Comparison

Factor	Stereo Vision	Motion Estimation
Accuracy	High	Lower
Cameras	Two	One
Hardware cost	Higher	Lower
System complexity	Higher	Lower
Deployment	More demanding	Easier
Best suited for	Accuracy-focused systems	Hardware-constrained systems

Stereo Vision

Stereo Vision uses two cameras positioned at different viewpoints. The difference between the views helps estimate the depth of objects.

Advantages:

- Provides accurate depth estimation.
- Suitable for applications requiring precise spatial information.
- Provides reliable depth when the cameras are properly calibrated.

Limitations:

- Requires two cameras.
- Increases hardware cost and system complexity.
- Requires proper camera calibration and physical setup.

Motion Estimation

Motion Estimation uses information from changes between frames captured by a single camera.

Advantages:

- Requires only one camera.
- Lower hardware cost.
- Easier to deploy in compact systems.
- Suitable when camera availability is limited.

Limitations:

- Provides less accurate depth estimation.
- Performance can depend on the quality and amount of motion.
- Less suitable for applications requiring highly precise depth.

Evaluation Under Hardware Constraints

If hardware resources are limited, using two cameras for Stereo Vision may not be practical. Motion Estimation is therefore more suitable because it can operate with a single camera and reduces hardware requirements.

However, if hardware limitations are not important and accurate depth estimation is the main requirement, Stereo Vision would be preferable.

Trade-Off

The main trade-off can be summarized as:

Stereo Vision → Higher accuracy + Higher hardware requirement

Motion Estimation → Lower accuracy + Lower hardware requirement

Thus, the choice depends on the application priority. Hardware-constrained systems should prioritize the simpler single-camera solution, while accuracy-critical systems can justify the additional hardware.

Conclusion

Based on the given data, Motion Estimation is preferable under hardware constraints because it requires only one camera and has lower system complexity. Although its accuracy is lower than Stereo Vision, this is a reasonable trade-off when hardware availability and cost are major constraints. If high depth accuracy is more important than hardware limitations, Stereo Vision is the better choice.

2. Model Generalization vs Overfitting

Performance:

Model	Training	Testing
A	98%	70%
B	90%	88%

Compare the models and evaluate which is more reliable for deployment. Justify based on generalization and risk factors.

Given Data

Model	Training Accuracy	Testing Accuracy
A	98%	70%
B	90%	88%

Comparison

Model A achieves a very high 98% training accuracy, but its testing accuracy drops significantly to 70%. This large difference indicates that the model has learned the training data too closely and does not perform equally well on unseen data.

Model B has 90% training accuracy and 88% testing accuracy. The small difference between the two values indicates that its performance remains consistent on unseen data.

Analysis

Model A – Overfitting

- Training accuracy = **98%**
- Testing accuracy = **70%**
- Accuracy gap = **28 percentage points**
- Performs very well on training data but poorly on new data.
- Indicates **overfitting**.
- Higher risk of unreliable predictions after deployment.

Model B – Better Generalization

- Training accuracy = **90%**
- Testing accuracy = **88%**
- Accuracy gap = **2 percentage points**
- Training and testing performance are very close.
- Indicates better generalization

- More likely to perform consistently on unseen real-world data.

Evaluation

The important factor is not simply the highest training accuracy. For deployment, testing performance and the gap between training and testing accuracy are more important.

Model A's 28-point gap shows that its learned patterns do not generalize effectively. Model B's 2-point gap demonstrates much better consistency.

Factor	Model A	Model B
Training accuracy	98%	90%
Testing accuracy	70%	88%
Train-test gap	28%	2%
Generalization	Poor	Good
Overfitting risk	High	Low
Deployment reliability	Lower	Higher

Conclusion

Model B is more reliable for deployment. Although Model A has the higher training accuracy, its large drop from 98% to 70% indicates severe overfitting. Model B maintains almost the same performance from 90% training to 88% testing, showing strong generalization and a lower risk of failure on unseen data. Therefore, Model B should be selected for real-world deployment because consistent testing performance is more valuable than exceptionally high training accuracy.

3. YOLO Performance across Scales

Detection results:

- **Large objects: 95%**
- **Medium objects: 85%**
- **Small objects: 60%**

Compare performance across scales and evaluate whether YOLO alone is sufficient for multi-scale detection tasks. Justify improvements.

Given Data

Object Size	Detection Accuracy
Large	95%
Medium	85%
Small	60%

Comparison and Analysis

The results show that YOLO performs best when detecting large objects and its accuracy decreases as object size becomes smaller.

- **Large objects – 95%:** YOLO provides excellent detection because large objects contain more visible features and occupy more pixels.
- **Medium objects – 85%:** Performance is still good, but some object details are reduced compared with large objects.
- **Small objects – 60%:** Detection accuracy drops considerably because small objects contain fewer pixels and their features can be difficult to distinguish from the background.

The difference between large and small object detection is 35 percentage points, showing a significant scale-related performance gap.

Evaluation

YOLO alone is not fully sufficient for reliable multi-scale detection, especially when small-object detection is important. Although its performance is strong for large and medium objects, the 60% accuracy for small objects indicates a major limitation.

Factor	Evaluation
Large objects	Excellent performance
Medium objects	Good performance
Small objects	Weak performance
Overall scale robustness	Limited
Main issue	Small-object detection

Improvements

The performance can be improved by:

1. **Multi-scale training** – Train the model using objects of different sizes so it learns scale variations.
2. **Higher-resolution input images** – Provides more pixels and useful features for small objects.
3. **Feature Pyramid Networks (FPN)** – Combines information from different feature levels to improve small-object detection.
4. **Data augmentation** – Scaling and cropping training images can provide more examples of small objects.
5. **Appropriate detection settings** – Adjusting model configuration and detection thresholds can improve detection performance for different object sizes.

Conclusion

YOLO achieves 95% accuracy for large objects and 85% for medium objects, but its performance falls to 60% for small objects. Therefore, YOLO alone is not sufficient when the application requires reliable detection across all object scales. Multi-scale training, higher-resolution inputs, feature-pyramid techniques, and suitable augmentation can improve its small-object detection capability.

4. Motion Model Selection for Animation

Two models:

- **Model A: smooth but physically inaccurate**
- **Model B: realistic but computationally intensive**

Compare the models and evaluate which should be used for animation systems under different constraints (real-time vs cinematic). Justify your decision.

Comparison of the Models

Factor	Model A	Model B
Motion	Smooth	Realistic
Physical accuracy	Low	High
Computational requirement	Low	High
Processing speed	Faster	Slower
Real-time suitability	High	Low
Cinematic suitability	Moderate	High
Main advantage	Speed and smoothness	Realistic motion

Analysis of Model A

Model A produces smooth motion, which is important when animation needs to respond immediately to user input or changing conditions. Its lower computational requirement allows the system to generate frames quickly.

However, the model is physically inaccurate. This means that although the movement may appear smooth, it may not correctly represent real-world physical behavior.

Therefore, Model A provides a good balance when speed and responsiveness are more important than physical realism.

Analysis of Model B

Model B focuses on producing realistic motion and better represents physical behavior. This makes it more appropriate when the visual quality and realism of animation are important.

The major limitation is that it is computationally intensive.

Therefore, Model B is more suitable when realism is the primary requirement and additional processing time is acceptable.

For **real-time animation**, Model A is the better choice.

Real-time systems require rapid processing because animation must respond continuously without noticeable delays. Model A's smooth motion and lower computational requirement allow faster frame generation.

Although its physical accuracy is lower, this limitation can be acceptable in applications where user interaction and responsiveness are more important.

Examples: interactive games, virtual environments, and real-time simulations.

Cinematic Animation

For cinematic animation, Model B is preferable.

Cinematic applications generally give greater importance to realistic movement and visual quality than immediate response. Since Model B produces more physically realistic motion, its higher computational cost can be accepted during offline rendering.

Examples: animated films, high-quality visual effects, and cinematic scenes.

Trade-Off Analysis

The two models represent a clear speed-versus-realism trade-off:

- **Model A:** Lower computation → faster processing → smooth animation → lower physical accuracy.
- **Model B:** Higher computation → slower processing → realistic animation → higher physical accuracy.

Thus, neither model is universally better. The appropriate choice depends on the application requirements.

Final Evaluation

Application Requirement	Recommended Model	Reason
Real-time response	Model A	Faster and computationally efficient
Interactive animation	Model A	Smooth motion with low delay
High physical realism	Model B	More realistic movement
Cinematic/offline rendering	Model B	Higher realism justifies computation

Conclusion

Model A should be selected for real-time animation because its smooth motion and lower computational requirement support fast and responsive performance. Model B should be selected for cinematic animation because its realistic motion provides better visual and physical accuracy, while its higher computational cost is acceptable for offline rendering.

5. Detection Model Comparison under Resource Constraints

Performance:

	Model Accuracy	Latency
A	85%	40 ms
B	92%	90 ms
C	90%	60 ms

Compare the models and evaluate which is optimal for real-time systems with moderate accuracy requirements. Justify your decision.

Given Data

Model	Accuracy	Latency
A	85%	40 ms
B	92%	90 ms
C	90%	60 ms

Comparison

Model A

- Accuracy: **85%**
- Latency: **40 ms**
- Fastest among all three models.
- Has the lowest accuracy.
- Suitable when response speed is the main priority.

Model B

- Accuracy: **92%**
- Latency: **90 ms**
- Provides the highest accuracy.
- Has the highest latency.
- Less suitable when strict real-time response is required.

Model C

- Accuracy: **90%**
- Latency: **60 ms**
- Provides better accuracy than Model A.
- Has considerably lower latency than Model B.
- Offers a good balance between accuracy and response time.

Analysis

The data shows a clear **accuracy–latency trade-off**.

- Model A reduces latency to **40 ms**, but accuracy is only **85%**.
- Model B improves accuracy to **92%**, but latency increases to **90 ms**.
- Model C achieves **90% accuracy at 60 ms**, providing a middle ground.

Since the question specifies **real-time systems with moderate accuracy requirements**, the system does not necessarily require the highest possible accuracy. It needs sufficiently accurate detection while maintaining fast response.

Evaluation

Requirement	Best Model
Lowest latency	A
Highest accuracy	B
Balanced accuracy and latency	C
Real-time with moderate accuracy	C

Although Model A is the fastest, its 85% accuracy may be unnecessarily low if better accuracy can be achieved without a large increase in latency. Model B provides the highest accuracy, but its 90 ms latency makes it less attractive for real-time operation.

Model C provides 90% accuracy with only 60 ms latency, making it the most balanced option.

Final Decision

Model C should be selected for the given real-time system.

Its 90% accuracy satisfies a moderate accuracy requirement, while its 60 ms latency provides faster response than Model B. Although Model A is faster, Model C provides a better accuracy–latency balance without the excessive delay of Model B.

Conclusion

For a real-time system with moderate accuracy requirements, Model C is the optimal choice because it provides a strong balance between detection accuracy and computational response time. Model A is preferable when minimum latency is critical, while Model B is suitable when accuracy is more important than real-time responsiveness. Therefore, Model C offers the most practical overall trade-off for the specified application.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Understanding of Data	25%	Accurately interprets all data patterns, trends, and relationships	Interprets data correctly with minor gaps	Partial understanding; misses key trends	Misinterprets or fails to understand data
Analysis	25%	Provides deep analysis with strong logical reasoning and justification	Provides reasonable analysis with some justification	Basic analysis with weak reasoning	No meaningful analysis provided
Application of Concepts	20%	Effectively applies relevant concepts/tools to interpret data accurately	Applies concepts with minor errors	Limited application; requires guidance	Unable to apply concepts to data
Conclusion	20%	Draws clear, meaningful conclusions with strong insights and implications	Provides valid conclusions with moderate insights	Conclusions are vague or weak	No clear or valid conclusions
Clarity	10%	Presents interpretation clearly using proper structure, visuals, and terminology	Generally clear with minor issues	Lacks clarity or proper structure	Poor presentation and difficult to understand